

Supplemental Materials: Cultural Cognition, Environmental Orientation, and Pro-Environmental Behavior

These supplemental materials report the single-orientation structural equation models — cultural cognition predicting the Connectedness to Nature Scale (CNS) and the New Ecological Paradigm (NEP) separately — the associated mediation analysis, and the full combined model estimated with the four separate worldview measures (hierarchy, egalitarianism, individualism, communitarianism). The main text reports the combined model with the four worldview measures collapsed into two composite groups, Hierarch-Individualist (HIER-INDIV) and Egalitarian-Communitarian (EGAL-COMM); the four-measure version here is the specification the composite-group model replaces.

Single-orientation structural models

Table 1: Model fit statistics for the CNS and NEP single-orientation structural equation models. Both models control for age, gender, race, education, and household income.

Model	N	df	Chi-sq	RMSEA	RMSEA 90% CI	CFI	TLI	SRMR
CNS	501	170	447.42	0.057	(0.051, 0.064)	0.906	0.877	0.077
NEP	501	170	608.59	0.072	(0.066, 0.078)	0.866	0.826	0.088

Serial chain vs. full model fit, composite worldview groups

The main text reports the full combined model (HIER-INDIV, EGAL-COMM composite groups) with direct paths added. Table S1a compares its fit against the strict serial-chain specification (no direct worldview → behavior paths) that motivated including those direct paths.

Figure 1: Measurement and structural model of cultural cognition, the Connectedness to Nature Scale, and public and private pro-environmental behaviors. Controlling for age, gender, race, education, and household income (paths not shown). Standardized coefficients shown; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

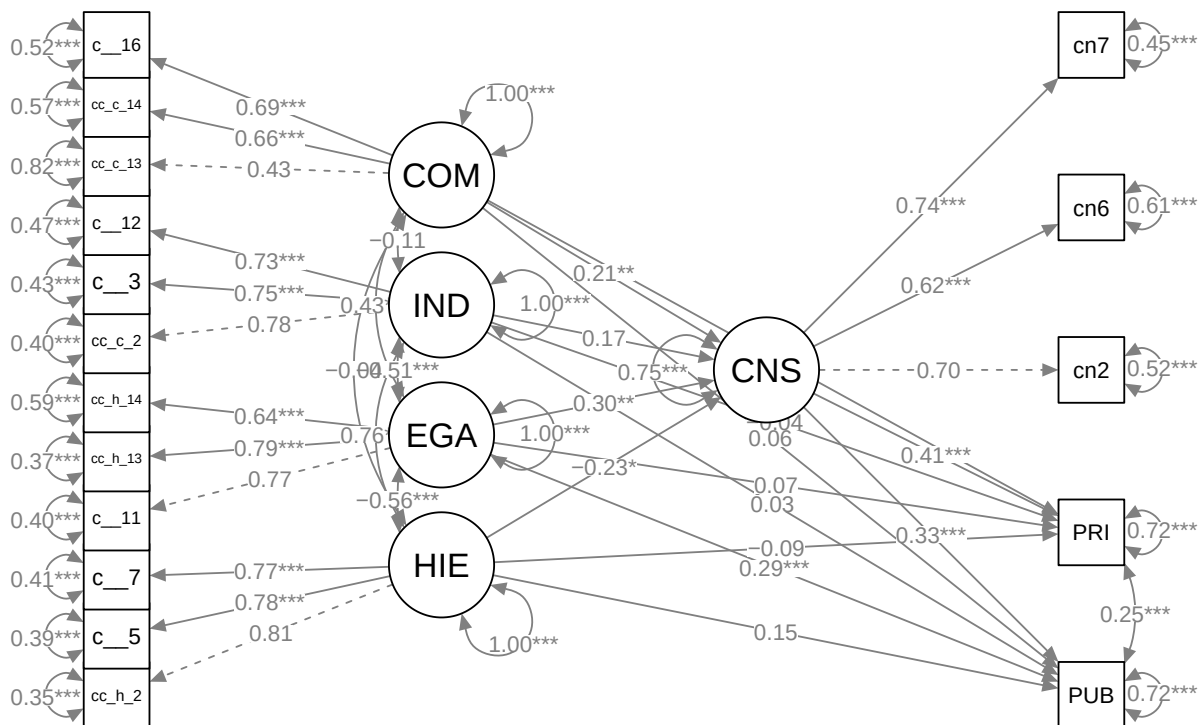


Figure 2: Measurement and structural model of cultural cognition, the New Ecological Paradigm, and public and private pro-environmental behaviors. Controlling for age, gender, race, education, and household income (paths not shown). Standardized coefficients shown; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

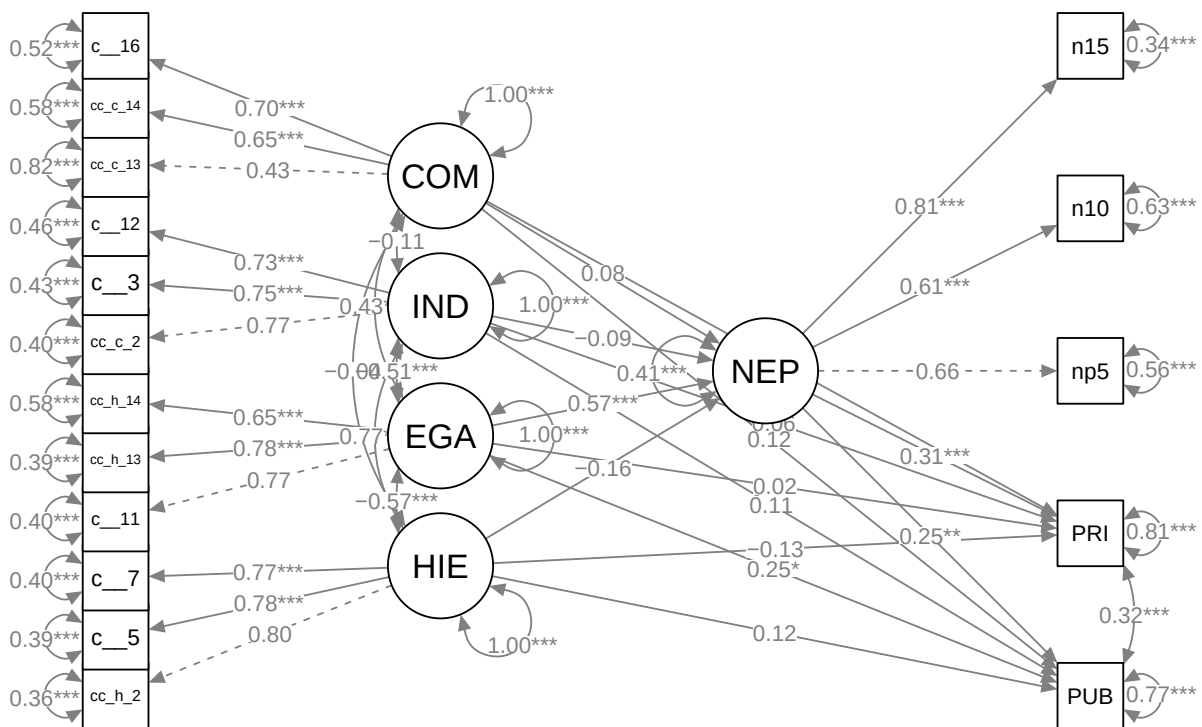


Table 2: Fit comparison of the strict serial chain versus the full combined model, composite worldview groups.

Model	Chi-sq	df	CFI	TLI	RMSEA	RMSEA 90% CI	SRMR	AIC	BIC
Serial chain (HI/EC)	171.87	45	0.881	0.815	0.075	(0.063, 0.087)	0.046	10,580.3	10,715.2
Full (HI/EC, + direct paths)	141.46	40	0.905	0.834	0.071	(0.059, 0.084)	0.039	10,559.8	10,715.9

Note: Both models control for age, gender, race, education, and household income. Nested χ^2 difference test (serial chain nested in full model, HIER-INDIV/EGAL-COMM groups): $\Delta\chi^2(5) = 30.41$, $p = 0.0000$.

Full combined model with the four separate worldview measures

This section reports the full combined model (cultural worldviews $\rightarrow$ NEP $\rightarrow$ CNS $\rightarrow$ behavior, with direct paths) estimated with the four separate worldview measures — hierarchy, egalitarianism, individualism, and communitarianism — as latent predictors. The main text reports the same model with these four measures collapsed into the two composite groups described above; this specification is the one the composite-group model replaces.

Table 3: Fit comparison of the strict serial chain versus the full combined model with direct paths added, four separate worldview measures. Both models control for age, gender, race, education, and household income.

Model	Chi-sq	df	CFI	TLI	RM-SEA	RMSEA 90% CI	SRMR	AIC	BIC
Serial chain	748.42	238	0.863	0.833	0.065	(0.060, 0.071)	0.085	25523.7	25827.3
Full (+ direct paths)	680.38	224	0.877	0.841	0.064	(0.058, 0.069)	0.081	25483.6	25846.3

Nested χ^2 difference test (serial chain nested in full model): $\Delta\chi^2(14) = 68.04$, $p = 0.0000$. The direct paths jointly improve fit.

Table 4: Standardized decomposition of cultural cognition effects on public and private behavior in the full combined model, four separate worldview measures, controlling for age, gender, race, education, and household income ($\dagger p < 0.10$, $* p < 0.05$, $** p < 0.01$, $*** p < 0.001$)

Predictor	Outcome	Direct	Serial (NEP $\rightarrow$ CNS)	Total indirect	Total
Egalitarianism	Public	0.213*	0.057*	0.176**	0.389***
Egalitarianism	Private	-0.023	0.071**	0.220***	0.197**
Communitarianism	Public	0.058	0.008	0.076**	0.134*
Communitarianism	Private	-0.141*	0.010	0.095**	-0.046

Table 4: Standardized decomposition of cultural cognition effects on public and private behavior in the full combined model, four separate worldview measures, controlling for age, gender, race, education, and household income († p<0.10, * p<0.05, ** p<0.01, *** p<0.001)

Predictor	Outcome	Direct	Serial (NEP→CNS)	Total indirect	Total
Hierarchy	Public	0.171†	-0.016	-0.093*	0.078
Hierarchy	Private	-0.066	-0.020	-0.116*	-0.182†
Individualism	Public	0.047	-0.009	0.038	0.086
Individualism	Private	-0.018	-0.011	0.048	0.029

Full combined model with measurement detail

Figure S4 shows the same four-worldview-measure full combined model with the full measurement model — the latent constructs and their indicators — included.

Mediation analysis

Table 5: Mediation analysis: standardized direct, indirect, and total effects of cultural cognition on public and private pro-environmental behavior, controlling for age, gender, race, education, and household income († p<0.10, * p<0.05, ** p<0.01, *** p<0.001)

Path	Outcome	Direct	Indirect	Total
Egalitarianism → CNS	Public	0.286***	0.099**	0.385***
Egalitarianism → CNS	Private	0.072	0.124**	0.197**
Communitarianism → CNS	Public	0.065	0.071*	0.135*
Communitarianism → CNS	Private	-0.136*	0.089*	-0.047
Egalitarianism → NEP	Public	0.249**	0.140**	0.389***
Egalitarianism → NEP	Private	0.022	0.177**	0.199**
Communitarianism → NEP	Public	0.116†	0.019	0.135*
Communitarianism → NEP	Private	-0.073	0.024	-0.049

Demographic control coefficients

Every model above and in the main text controls for age, gender (male), race (white), education, and household income, but these coefficients are omitted from the path diagrams and decomposition tables for readability. Table S6 reports the standardized control coefficients from the full combined model reported in the main text (HIER-INDIV and EGAL-COMM in place of the four separate worldview measures; main-text Table 6).

Figure 3: Structural model of the full combined SEM with the four separate worldview measures: cultural cognition → New Ecological Paradigm → Connectedness to Nature → public and private pro-environmental behavior, with direct paths. Controlling for age, gender, race, education, and household income (paths not shown). The measurement (indicator) portion is omitted for clarity; standardized coefficients shown; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

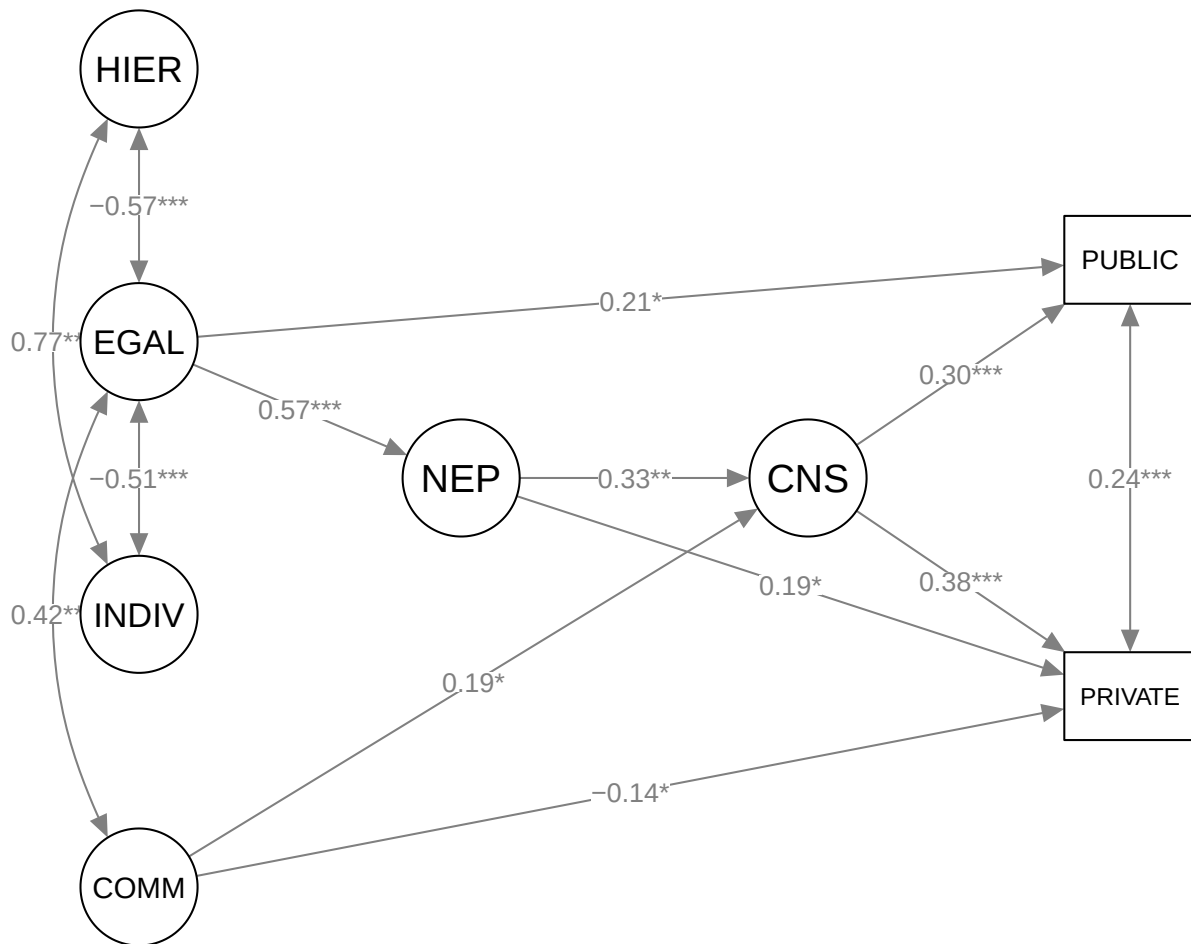


Figure 4: Full combined model including the measurement model (latent constructs and their indicators) and the structural paths: cultural cognition → New Ecological Paradigm → Connectedness to Nature → public and private pro-environmental behavior, with direct paths. Controlling for age, gender, race, education, and household income (paths not shown). Standardized coefficients shown; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

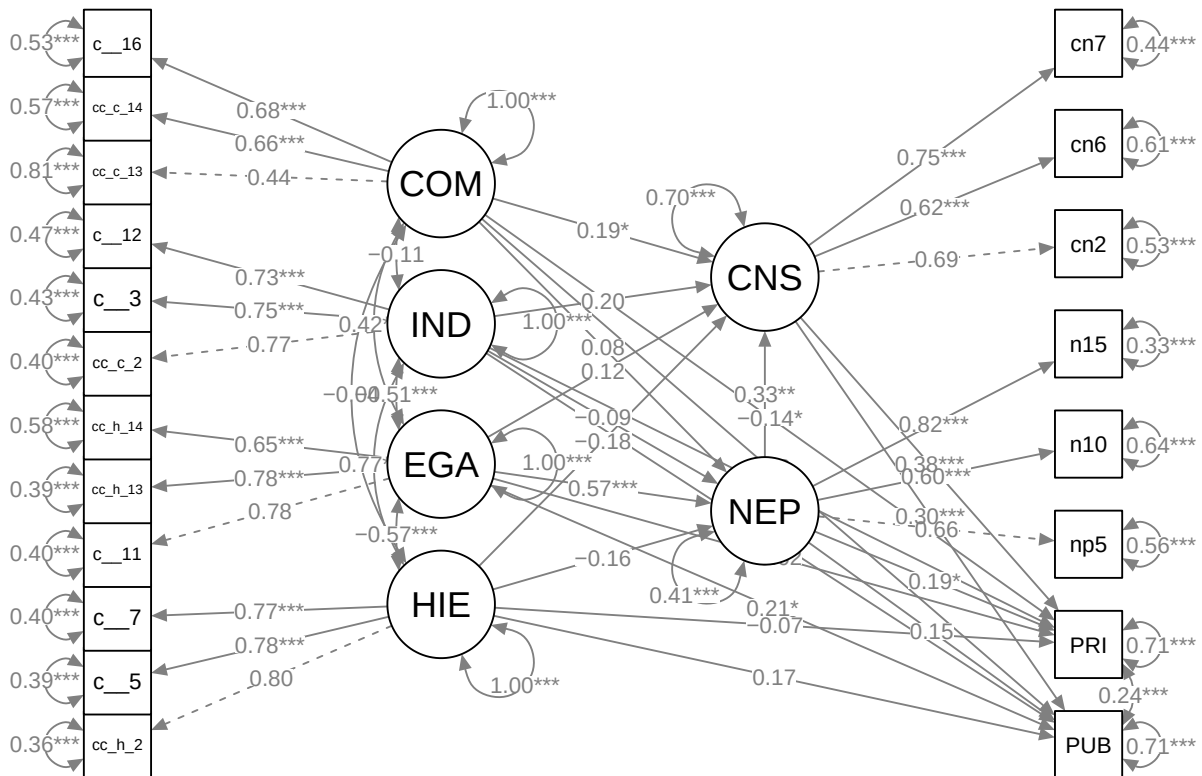


Table 6: Standardized demographic control coefficients from the full combined model reported in the main text (HIER-INDIV/EGAL-COMM), by outcome († $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$)

Control	NEP	CNS	Behavior
Age	-0.039	0.090†	-0.059
Male	-0.072	0.010	-0.026
White	-0.030	0.026	0.029
Education	-0.078	0.019	0.071
Household income	-0.066	-0.045	0.134**

References